Kushagra Garg

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LinkedIn/arXiv/GoogleScholar
Webpage/GitHub/X

Education

University of Toronto — Toronto, Canada
Ph.D. in Computer Science — Sep 2026 - Present
Department of Computer Science
- Supervisor: Nathan Wiebe

International Institute of Information Technology — Hyderabad, India
M.S. in Computational Natural Science — July 2019 - Dec 2025
- Advisors: Shantanav Chakraborty, Subhadip Mitra
- Research area: Quantum Algorithms
- GPA: 9.33/10

B.Tech in Computer Science — July 2019 - Dec 2025
- GPA: 8.85/10

Publications

- **K. Garg**, Z. Ahmed, S. Mitra, S. Chakraborty. Simulating quantum collision models with Hamiltonian simulations using early fault-tolerant quantum computers. *Physical Review A*, vol. 112, no. 2, Aug. 2025.
- **K. Garg**, Z. Ahmed, A. Thomasen. Entanglement determination with a deep multiscale entanglement renormalization ansatz. *Physical Review A*, vol. 111, no. 4, Apr. 2025.
- Z. Ahmed, A. Chaudhuri, K. S. S. Grover, A. Rao, **K. Garg**, P. Malhotra. Classifying CELESTE as NP-complete. *arXiv:2012.07678*, 2020.
- M.S. Thesis: Early Fault-Tolerant Quantum Algorithms for Open Quantum Systems

Research Experience

Fujitsu Research — Bangalore, India
Research Scientist — July 2024 - August 2026
- Designed a Lindbladian simulation and Gibbs state preparation algorithm for early fault-tolerant quantum computers, requiring only a single ancilla qubit.
- Achieved exponential circuit-depth reduction through polynomial interpolation techniques.
- Validated the end-to-end method, demonstrating up to 10,000× precision improvement over existing approaches at comparable circuit depths.
- Co-developing a quantum resource estimation package to enable comparisons across algorithms, error-correction schemes, and device architectures.

Microsoft Research — Bangalore, India
Research Intern — Feb 2024 - July 2024
- Explored the application of multi-party computation for secure inference of large language models, focusing on OT-based protocols.
- Leveraged symmetry properties of the fundamental neural network operations (ReLU, GeLU, LayerNorm) to optimize secure inference protocols, reducing computational and communication costs by more than 50%.

QunaSys Inc. Tokyo, Japan
Research Intern June 2022 - Jan 2024

- Developed qubit-efficient entanglement estimation methods, enabling simulations of larger quantum states on limited hardware.
- Used ZX-calculus to show that MERA tensor-network ansätze avoid barren plateaus, making them more reliable for variational algorithms.
- Contributed hardware-tailored diagonalization algorithms to the Quri-SDK package, enabling significant reduction in measurement circuit depth.

IIIT Hyderabad Hyderabad, India
Student Researcher and Teaching Assistant July 2019 - July 2025

- Developed early fault-tolerant quantum algorithms for simulating quantum collision models to characterize both Markovian and non-Markovian open quantum systems.
- Researched quantum algorithmic approaches based on quantum walks for parton shower simulation in high-energy physics.
- Served as Teaching Assistant for Automata Theory and Statistical Mechanics, conducting weekly tutorials, designing problem sets, and grading assignments and exams.

Talks

- Early Fault-Tolerant Simulation of Open quantum Systems. Presented at *Quantum Meets*.

Awards

- Dean's List Academic Award 2023
- Winner of Quantum Chemistry Challenge Qhack 2022
- KVPY Scholarship 2019

Academic Services

Reviewer for: *Quantum Journal*, *AQIS 2025*.
Student Moderator: *ACM CompEd 2023*, *NQSTS 2021*.
Club Coordinator: The Art Society, Astronomy Club.

Skills

Languages: English (*TOEFL*: 113/120), Hindi.
Programming: Python, C/C++, MATLAB, Go.
Miscellaneous: LaTeX, Git, Qiskit.